Daniel Flynn

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Education

Northeastern University, Boston, MA — December 2025

Bachelor of Science in Computer Engineering and Computer Science

GPA: 3.59 (Cum Laude)

Relevant Coursework: Computer Architecture, Natural Language Processing, Software Engineering, Digital Design, Capstone, Linear Systems, Artificial Intelligence, Random Phenomena in EE, Computer Systems, Object-Oriented Design, Algorithms and Data, Logic and Computation, Circuits and Signals, Fundamentals of Networks, Embedded Design

Skills

Languages: Python, C/C++, JavaScript/TypeScript, Java, Assembly (x86/ARM/MIPS), Verilog, SQL, LaTeX, Bash
Software: Git, GitHub Actions, CI/CD, Docker, Kubernetes, Netlify, YAML, CMake, Makefile, SystemD, Ansible, SSH
Hardware and Embedded: Linux, STM32, ESP32, Raspberry Pi, Vivado, FPGAs, UART, SPI, I2C, JTAG, DFU, Qt
Web Development: Node, React, Vite, HTML/CSS, MongoDB, RESTful APIs, Sockets, OpenAPI, Jest, Cypress
Machine Learning: NumPy, PyTorch, TensorFlow, Hugging Face, Transformers, NLP, Deep Learning, OpenCV

Experience

NK Labs

Embedded Software Engineering Co-op — January 2025 – June 2025

- Developed software and firmware for a PCR machine using an STM32 microcontroller integrated with a TI AM62 SoC.
- Wrote C++ front-end code with Qt for menus, buttons, experiment graphs/files, and back-end communication over sockets.
- Implemented experiment timers, estimated time calculation, temperature ramping, and event handling on Python back-end.
- Created JavaScript web app to control the PCR machine with socket events by connecting to the device's IP in a browser.
- Built a custom Linux image with Buildroot, integrating front-end and back-end daemons for the PCR machine.

Schneider Electric

Software Engineering Co-op — January 2024 – June 2024

- Wrote C++ code handling server/client communication between edge devices using OPC UA and MQTT protocols.
- Deployed edge computing middleware for industrial automation, orchestrating services with Docker and Kubernetes.
- Cross-compiled edge computing middleware for multiple CPU architectures and integrated WebAssembly modules.
- Contributed to DevOps processes, CI/CD pipeline, and network management using GitHub Actions and Ansible.

Depuy Synthes Mitek Sports Medicine (Johnson and Johnson)

Software Engineering Co-op — January 2023 – June 2023

- Developed software and firmware for the FMS VUE saline and fluid management pump, utilizing a monochrome dot-matrix LCD for the front-end, and IBM Rhapsody for UML design, code organization, simulation, and validation on the back-end.
- Wrote C code for FMS VUE firmware and Python code to parse information from SystemD logs using journalctl.
- Used Arm Keil compiler and JTAG debugger to program flash memory, debug firmware code, and create release versions.
- Implemented auto calibration mode for FMS VUE pressure compensation based on elevation with laboratory testing.

Selected Projects

ARM LEGv8 CPU Implementation with Schematic Compiler — November 2025 – December 2025

- Verilog implementation and testbench for a 5-stage pipelined CPU with subset of the ARMv8 instruction set, 64-bit registers, ALU, register file, instruction decoder, program counter, control signals, and branching.
- Python schematic compiler that parses verilog code to generate a directed graph of components using DFS and Graphviz.
- CPU featured in the Northeastern COE Newsletter; Schematic compiler was built as a personal project outside of class.

The Cool Bike Video Game and Engine — January 2026 – Present

- Created an OpenGL rendering pipeline using models, meshes, shaders, and glm linear algebra for camera transformations.
- Implemented collision and gravity physics using hitbox vectors generated from glb 3d models and quaternions for rotations.
- Wrote unit tests for transform logic, camera positioning, and hitbox detection using C++ gtest library for validation.

Embedded Environmental Monitoring and Control System — April 2026 – May 2026

- Integrated circuit that uses an 8 relay module, AHT21, PCF8574, STC89, and W5500 to control heaters and humidifiers.
- MCU firmware that takes sensor measurements, sends UDP data, receives TCP settings, and saves settings to EEPROM.
- Qt app that allows users to view measurement data and configure temperature/humidity thresholds for turning on relays.
- Bit banged I2C and SPI drivers written in assembly for 8051 architecture. Firmware written in C with SDCC 8051 header.